

Product Line Profitability & Margin Performance Analysis for Nassau Candy Distributor

Final Internship Research Report

Submitted To: Unified Mentor

Submitted By: Mahek Pandey

BCA with Artificial Intelligence

V.T. Poddar College

Certificate

This project report entitled 'Product Line Profitability & Margin Performance Analysis for Nassau Candy Distributor' has been prepared by Mahek Pandey as part of the Unified Mentor Internship. The work presented is original and completed for academic learning purposes.

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Acknowledgement

I sincerely thank Unified Mentor for providing this internship opportunity. I am grateful to my mentors, faculty members and V.T. Poddar College for their continuous guidance and encouragement during the completion of this project.

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Abstract

This report presents a Business Intelligence solution for evaluating product profitability, gross margin, regional performance and division-wise performance using Python, Pandas, Plotly and Streamlit.

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Introduction

Organizations require profitability analysis in addition to revenue analysis. This project transforms raw sales transactions into actionable business insights through interactive dashboards.

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Problem Statement

Management lacks clear visibility into product profitability, margin risk and division performance, making strategic decisions difficult.

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Objectives

Analyze sales, cost, gross profit, gross margin, division performance, regional performance, Pareto analysis, cost diagnostics and build an interactive dashboard.

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Literature Review

Business Intelligence dashboards support data-driven decision making by converting operational data into visual insights. Previous studies emphasize KPI monitoring, profitability analysis, margin optimization and dashboard-driven management.

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Dataset Description

The dataset contains order, customer, product, division, region, sales, cost, units and gross profit information suitable for profitability analysis.

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Methodology

The methodology includes data collection, preprocessing, missing value validation, feature engineering, exploratory data analysis, KPI generation, visualization and dashboard development.

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Data Cleaning

Duplicate records were checked, date fields converted, numerical values validated and new profitability metrics engineered.

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Feature Engineering

Gross Margin Percentage, Profit per Unit, Revenue Contribution and Profit Contribution were calculated for deeper analysis.

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Exploratory Data Analysis

EDA included revenue distribution, profit distribution, cost analysis, division comparison, regional analysis, Pareto analysis, correlation analysis and risk identification.

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Dashboard Development

A Streamlit dashboard was developed with KPI cards, filters, product profitability analysis, division performance, regional analytics, cost diagnostics and downloadable reports.

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Results and Discussion

The analysis shows that a limited number of products contribute to the majority of profit. Some products generate high revenue but relatively low margin, indicating opportunities for pricing optimization and cost reduction.

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Business Insights

High-margin products should receive greater marketing investment. Low-margin products should be reviewed for pricing and manufacturing cost optimization. Regional and division performance varies significantly.

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Recommendations

Optimize pricing, reduce manufacturing costs, prioritize profitable products, improve inventory planning and continuously monitor KPIs.

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Conclusion

The project demonstrates the effectiveness of Business Intelligence for improving strategic decision making and profitability through interactive analytics.

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Future Scope

Future enhancements include demand forecasting, machine learning, customer segmentation, supplier analytics and automated reporting.

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References

Python Documentation

Pandas Documentation

Plotly Documentation

Streamlit Documentation